



UNIVERSITY OF RAJSHAHI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSE4261 NEURAL NETWORKS AND DEEP LEARNING

Assignment-11

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July 25, 2025

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Abstract

This report investigates the critical role of loss functions in two distinct deep learning applications. The first part of the report addresses the question: **How do three different face verifiers compare when trained using Binary Cross-Entropy loss, Contrastive loss, and Triplet loss, respectively?** This analysis delves into how each loss function structures the embedding space for the 1:1 face matching task. The second part of the report answers the question: **What is the effect of using Binary Cross-Entropy (BCE) versus Mean Squared Error (MSE) as a reconstruction loss during the training of a Variational Autoencoder (VAE)?** This section explores how the choice of reconstruction loss influences the visual characteristics and quality of the images generated by the VAE. Each loss function is presented mathematically and its conceptual impact on model training and performance is discussed.

1 Comparing Loss Functions for Face Verification

Face verification is a "1:1" matching problem: given two images, determine if they belong to the same person. The core of modern face verification systems is a deep neural network that maps a face image to a low-dimensional feature vector, or "embedding." The choice of loss function is critical for training the network to produce an embedding space where distances directly correspond to facial similarity. This report compares three common approaches.

1.1 Method 1: Binary Cross-Entropy Loss

This approach treats face verification as a simple binary classification problem. A Siamese network architecture is typically used, where two images are passed through identical CNNs to get their respective embeddings. These embeddings are then combined (e.g., by taking the absolute difference or distance) and fed into a final dense layer that predicts whether the pair is "same" (1) or "different" (0).

Loss Function

The Binary Cross-Entropy (BCE) loss is used to train the final classification head. The formula is:

$$L_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

where y_i is the true label (1 for a genuine pair, 0 for an imposter pair) and $\hat{y}_i$ is the model's predicted probability that the pair is genuine.

Result



Figure 1: Training loss of BCE loss function

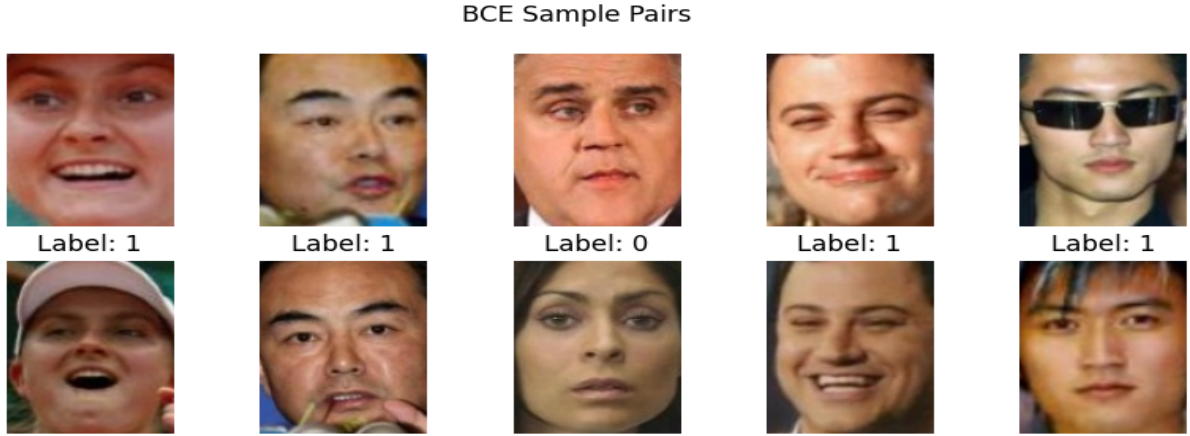


Figure 2: Face verification and loss curve for BCE loss function

1.2 Method 2: Contrastive Loss

Unlike BCE, contrastive loss works directly on the distances in the embedding space. Its goal is to minimize the distance between embeddings of genuine pairs (pulling them together) while ensuring the distance between embeddings of imposter pairs is above a certain margin (pushing them apart).

Loss Function

Contrastive loss takes a pair of embeddings and their label ($y = 1$ for genuine, $y = 0$ for imposter) as input. The formula is:

$$L_{\text{contrastive}} = y \cdot \frac{1}{2} D_w^2 + (1 - y) \cdot \frac{1}{2} \max(0, m - D_w)^2 \quad (2)$$

where D_w is the Euclidean distance between the two embeddings, and m is the margin, a hyperparameter that defines the minimum desired distance for imposter pairs.

Result

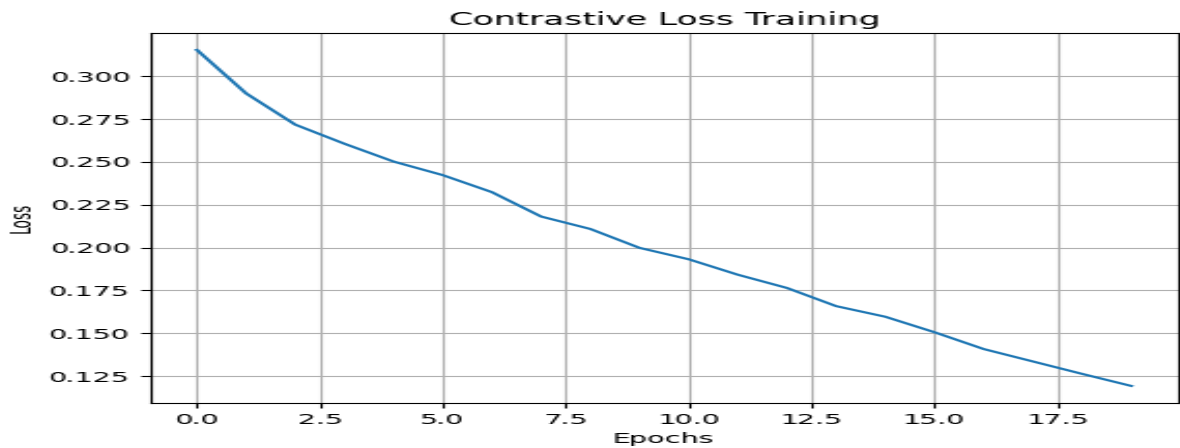


Figure 3: Training loss of Contrastive loss function

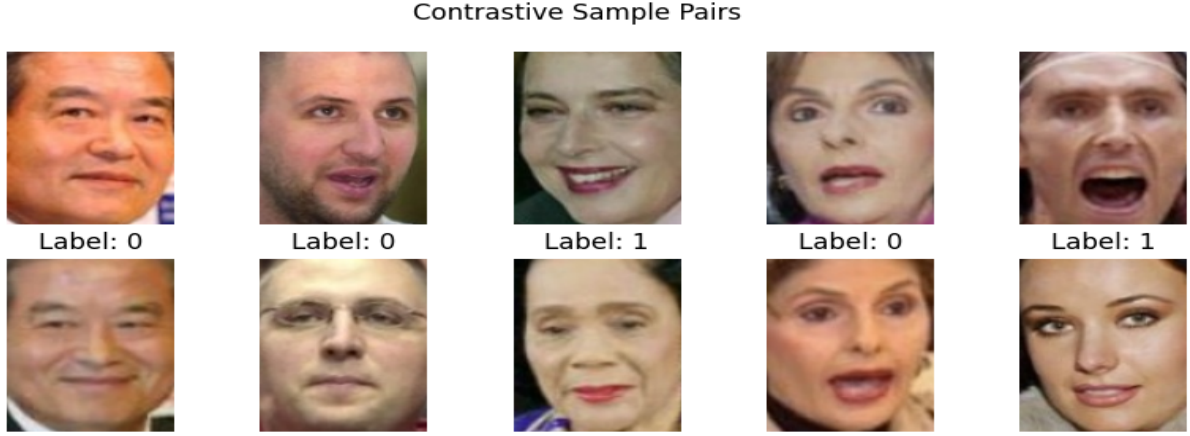


Figure 4: Face verification and loss curve for Contrastive loss function

1.3 Method 3: Triplet Loss

Triplet loss is often considered a more powerful metric learning approach. Instead of learning from pairs, it learns from triplets of images:

- **Anchor (A):** A baseline image of a person.
- **Positive (P):** A different image of the same person.
- **Negative (N):** An image of a different person.

The objective is to ensure the distance between the Anchor and the Positive is smaller than the distance between the Anchor and the Negative, by at least a certain margin (α).

Loss Function

The triplet loss function is defined as:

$$L_{\text{triplet}} = \max(\|f(A) - f(P)\|_2^2 - \|f(A) - f(N)\|_2^2 + \alpha, 0) \quad (3)$$

where $f(\cdot)$ is the embedding function (the CNN), and α is the margin. The loss is only non-zero if the negative sample is "too close" to the anchor relative to the positive sample.

Result

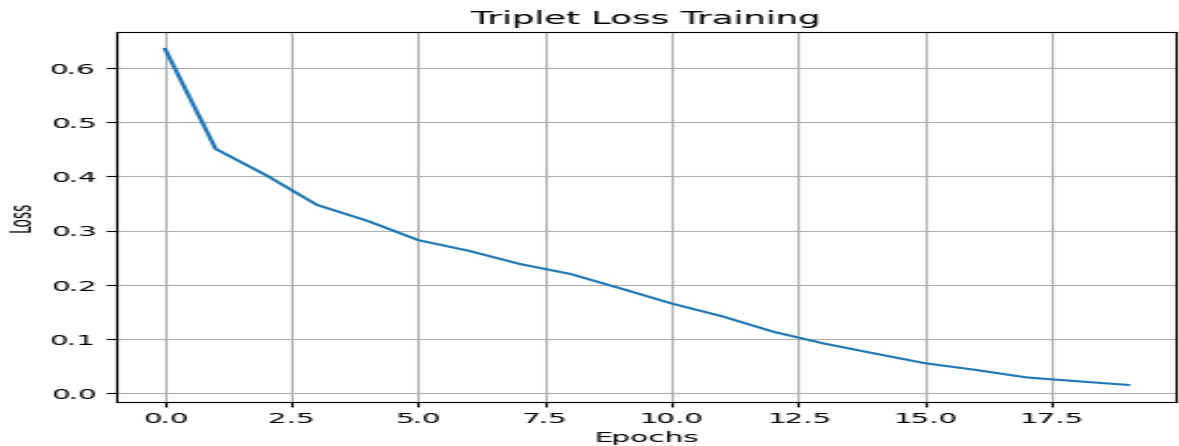


Figure 5: Training loss of Triplet loss function



Figure 6: Face verification and loss curve for Triplet loss function

1.4 Comparison and Analysis

- **Binary Cross-Entropy** is the simplest approach but is an indirect way of learning an embedding space. It focuses on the classification outcome rather than the quality of the feature space itself.
- **Contrastive Loss** is a significant improvement as it directly optimizes the embedding space distances. However, it only considers pairs in isolation.
- **Triplet Loss** is often the most effective. By considering relative distances and enforcing a margin, it creates a more robust and well-structured embedding space. Its main challenge is the need for "triplet mining"—finding triplets that are effective for training (i.e., where the negative is not already far away).

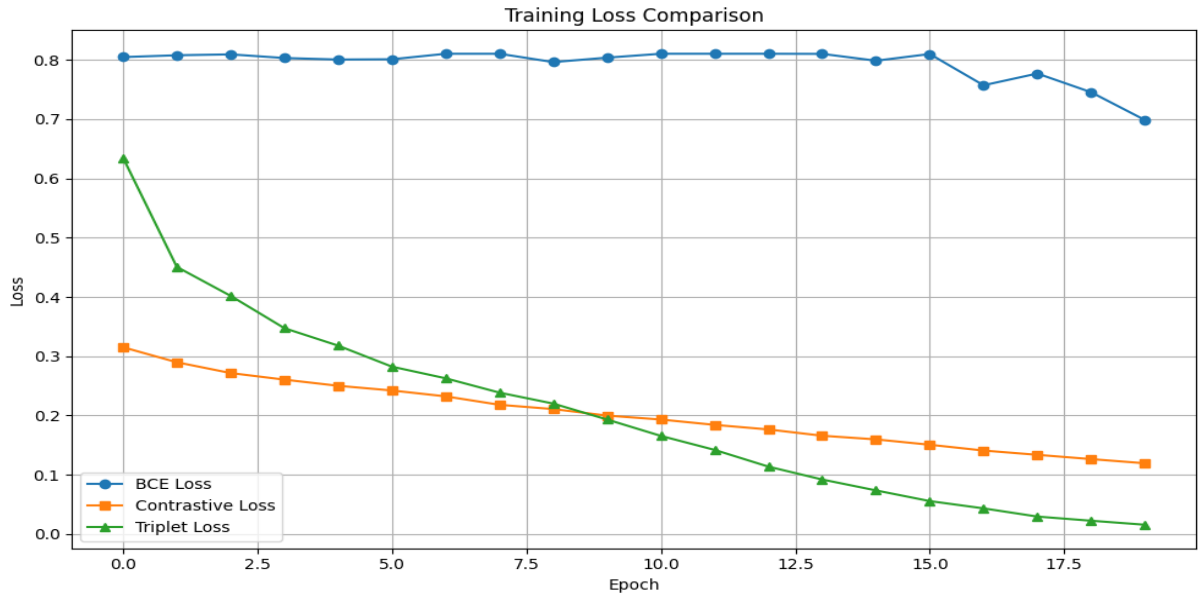


Figure 7: All loss curve comparison

2 Effect of Reconstruction Loss in a VAE

A Variational Autoencoder (VAE) is a generative model that learns a compressed latent representation of data. Its loss function has two components: a **reconstruction loss** that forces the VAE to accurately recreate the input, and the **KL divergence**, which regularizes the latent space. This report compares two common choices for the reconstruction loss term.

2.1 The VAE Loss Function

The overall VAE loss is a sum of the reconstruction term and the KL divergence term:

$$L_{\text{VAE}} = L_{\text{reconstruction}} + \beta \cdot D_{\text{KL}}(\mathcal{N}(\mu, \sigma) || \mathcal{N}(0, 1)) \quad (4)$$

The choice of $L_{\text{reconstruction}}$ depends on the assumed probability distribution of the input data.

2.2 Method 1: Binary Cross-Entropy (BCE) as Reconstruction Loss

BCE is the appropriate choice when the input data is normalized to a range of $[0, 1]$ and can be modeled as a Bernoulli distribution (i.e., each pixel is a probability). This is common for datasets like MNIST.

Loss Function

The reconstruction loss is the sum of the BCE loss over all pixels:

$$L_{\text{BCE}} = - \sum_{i=1}^D [x_i \log(\hat{x}_i) + (1 - x_i) \log(1 - \hat{x}_i)] \quad (5)$$

where D is the number of pixels, x_i is the value of the original pixel, and $\hat{x}_i$ is the value of the reconstructed pixel.

Effect on VAE Output

Using BCE encourages the decoder to output values that are close to 0 or 1. As it is a measure of distributional similarity rather than exact value matching, it tends to produce reconstructions that are perceptually plausible but can be slightly **blurry or soft**. It is less sensitive to small pixel-wise errors compared to MSE.

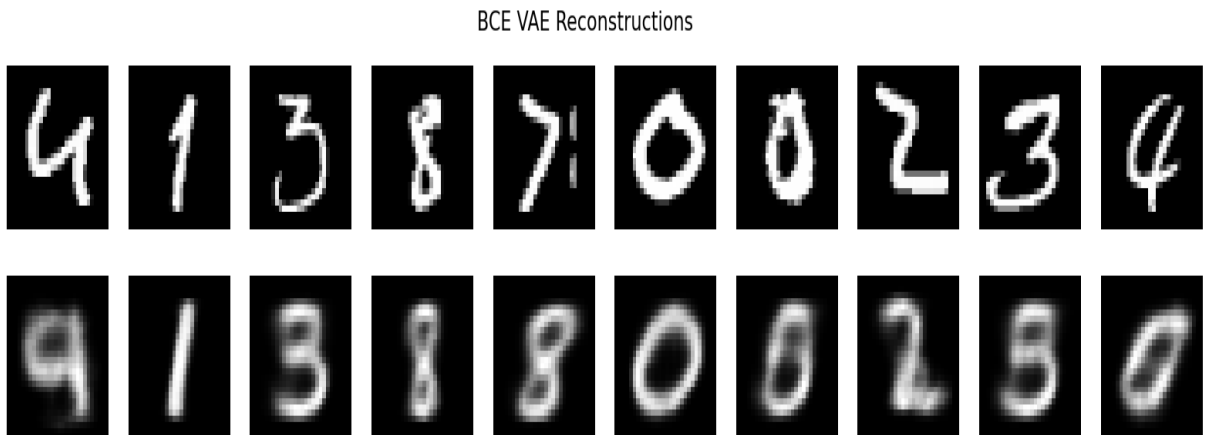


Figure 8: Image reconstruction using VAE with BCE loss function

2.3 Method 2: Mean Squared Error (MSE) as Reconstruction Loss

MSE is used when the input data is assumed to follow a Gaussian distribution. This is suitable for images with continuous pixel values that are not necessarily restricted to $[0, 1]$.

Loss Function

The MSE loss, also known as L2 loss, measures the average squared difference between the original and reconstructed pixels:

$$L_{\text{MSE}} = \frac{1}{D} \sum_{i=1}^D (x_i - \hat{x}_i)^2 \quad (6)$$

Effect on VAE Output

Because MSE heavily penalizes large errors, it pushes the decoder to produce pixel values that are very close to the original. This can result in **sharper reconstructions** on average. However, it can also lead to a "regression to the mean" effect, where the model produces less vibrant or grayish images, as predicting the average color is a safe way to minimize squared error.

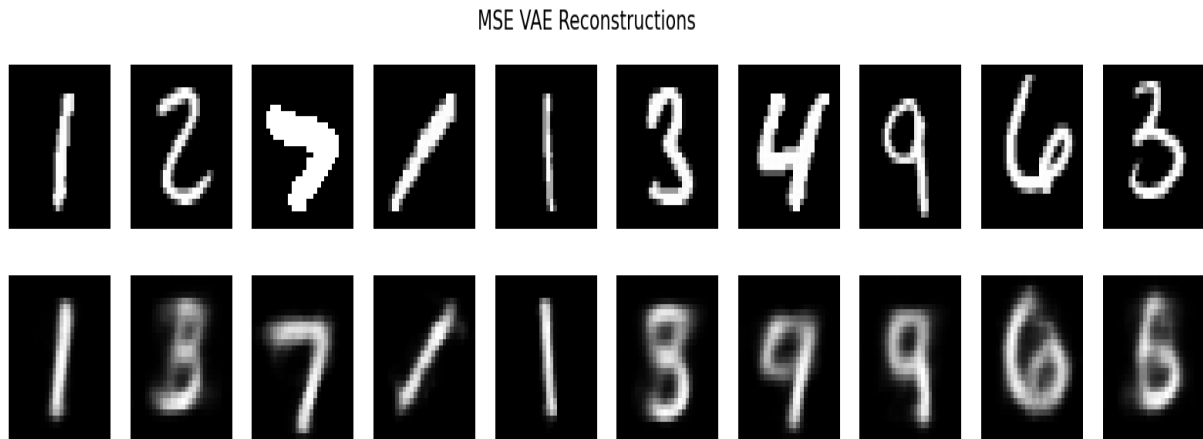


Figure 9: Image reconstruction using VAE with MSE loss function

2.4 Comparison and Analysis

The choice between BCE and MSE is not arbitrary; it is a modeling decision based on the data's nature.

- Use **BCE** for data that is normalized to $[0, 1]$ and can be treated as probabilities (e.g., black and white images). This often yields softer but more stable results.
- Use **MSE** for data with continuous values (e.g., color images normalized to $[-1, 1]$). This can yield sharper images but may lack perceptual vibrancy compared to more advanced loss functions like perceptual loss.

Ultimately, the visual quality of the generated samples depends heavily on this choice.

3 Conclusion

This report has analyzed the critical role of the loss function in two distinct deep learning domains. In face verification, moving from an indirect method like Binary Cross-Entropy to

direct metric learning with Contrastive and Triplet Loss allows a model to build a more powerful and semantically meaningful embedding space. In generative modeling with VAEs, the choice between Binary Cross-Entropy and Mean Squared Error as the reconstruction loss directly impacts the visual characteristics of the generated output, with BCE often producing softer images and MSE producing sharper ones. In both cases, the loss function is not merely a final step but a core component that defines the model's objective and shapes its final performance.

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